NGUYEN DUC TOAN

AI ENGINEER | DATA SCIENTIST

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SUMMARY

AI Engineer with experience in chatbot development and NLP applications, specializing in intelligent conversational systems, LLM integration, prompt optimization, and user experience improvement. Strong foundation in Data Science, Machine Learning, data analysis, and model evaluation. Capable of delivering end-to-end AI solutions, from data processing and model development to API integration and production deployment.

TECHNICAL SKILLS

AI / Chatbot / NLP

- Developed chatbots using LLMs, Prompt Engineering, and RAG.
- Built semantic search and knowledge retrieval systems with embeddings and vector databases.
- Integrated chatbot solutions with APIs, databases, and internal platforms.
- Optimized prompts and retrieval pipelines to improve answer relevance and reduce hallucinations.
- Implemented fallback mechanisms and guardrails for more reliable responses.

Data Science / Machine Learning

- Performed data preprocessing, cleaning, feature engineering, and exploratory data analysis (EDA).
- Built and evaluated machine learning models for classification, regression, and clustering.
- Applied statistical analysis and visualization to support data-driven decision-making.
- Optimized model performance using threshold tuning, ensemble methods, and metric-based evaluation.
- Measured results using Accuracy, Precision, Recall, F1-score, ROC-AUC, MSE, and R^2.

Tools / Technologies

- Python, SQL, Pandas, NumPy, Scikit-learn.
- XGBoost, LightGBM, LangChain, FastAPI, Streamlit, ZenML.
- TiDB Vector, MinIO, LangSmith.
- Git, GitHub, Docker.

WORK EXPERIENCE

AI Engineer

11/2025 - 09/2026

SOFT WORLD VIETNAM COMPANY LIMITED

- Built RAG pipelines with embeddings and vector search to enable knowledge retrieval from internal documents.
- Designed and optimized prompts for LLM-based chatbot systems to improve response accuracy and naturalness.
- Integrated chatbots with APIs, databases, and business systems to provide real-time responses.
- Monitored conversation logs, analyzed user behavior, and improved dialogue flows based on real usage data.

ACTIVITIES AND ACHIEVEMENTS

- Participated in SUMMER CODE CAMP 2024 – Faculty of IT, HCMUTE.
- Participated in Samsung Innovation Campus (SIC) 2025 – AI Bootcamp (6-month program).
- Commended as an Outstanding Student in Academic Performance - Academic Year 2023–2024.
- Partcipipated in AWS First Cloud Journey (FCJ) Session 2. .

EDUCATION

Chu Van An High School **2020 - 2023**

Specialized Mathematics-Informatics Class

Ho Chi Minh City University of Technology and Education **2023-Present**

Data Engineer

PROJECTS

FLOWASSIST ChatBot with LLM and RAG

Developed an AI chatbot for automated question answering over business documents and internal knowledge using a **Retrieval-Augmented Generation (RAG)** architecture. Built a semantic retrieval pipeline with **multilingual embeddings** and **TiDB Vector**, combined with **Llama 3.3 70B (via Groq API)** for accurate, context-aware responses.

Optimized performance through chunking strategies, retrieval threshold tuning, and prompt engineering to improve answer quality. Built backend services with **FastAPI**, integrated **Redis for caching, MinIO for document storage**, and **Alembic for database migrations**. Deployed and managed the system on **AWS EC2**.

Github: https://github.com/GrayNguyen-Data/FlowAssist-Chatbot

Skills & Technologies

Python, LLM, RAG, LangChain, Groq API, Llama 3.3 70B, TiDB Vector, Vector Search, Prompt Engineering, MinIO, LangSmith, FastAPI/Backend API, NLP, Redis, EC2.

MLOps Pipeline for House Price Prediction

Developed an end-to-end **MLOps pipeline** for house price prediction using **ZenML**, covering data ingestion, preprocessing, feature engineering, model training, evaluation, and model versioning. Built and compared multiple regression models including **Ridge Regression, Random Forest, XGBoost, and LightGBM**, then improved predictive performance with a stacking ensemble approach. Logged metrics and metadata into the ZenML Model Registry and built a **Streamlit app** for interactive model testing. Achieved best model performance of **$R^2 \approx 0.9421$ and MSE ≈ 0.0081** on the Ames Housing dataset.

Github: https://github.com/GrayNguyen-Data/MLOps-Pipeline-for-House-Price-Prediction

Skills & Technologies

Python, ZenML, MLOps, Machine Learning, Regression, Stacking Ensemble, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Streamlit, Model Registry, Feature Engineering.

CUSTOMER CHURN PREDICTION

Developed an end-to-end customer churn prediction pipeline on a telecom dataset of 7,043 customers using **Random Forest, XGBoost, and LightGBM.** Built Random Forest as a baseline model and improved churn detection with boosting-based approaches under class imbalance. Evaluated model performance using **ROC-AUC, Precision, Recall, and F1-score** with a focus on business impact, and applied probability threshold tuning to improve churn detection. Achieved high recall of approximately 0.86 for churn customers, while **XGBoost and LightGBM** delivered stronger overall performance with **ROC-AUC around 0.83.**

Github: https://github.com/GrayNguyen-Data/CUSTOMER-CHURN-PREDICTION

Skills & Technologies

Python, Random Forest, XGBoost, LightGBM, Threshold Tuning, ROC-AUC, Precision, Recall, F1-score, Pandas, NumPy, Scikit-learn, EC2.

REGIONAL RETAIL ANALYSIS DASHBOARD | Tableau BI Project

Developed an interactive Tableau dashboard for analyzing regional sales, profit, and order performance. Implemented KPI tracking, daily sales/profit trend analysis, and Top 5 profitable sub-category visualization with dynamic region-based filtering.

Github:https://github.com/GrayNguyen-Data/REGIONAL-RETAIL-SALES-REVIEW

Skills & Technologies: Tableau Desktop, Data Visualization, Business Intelligence (BI), KPI Analytics, Retail Analytics, Dashboard Development